Unit 3 Challenge Exam

Math 9E Parts 1–3 · Lessons 3.1–3.4 13 questions · 66 marks Name: _____

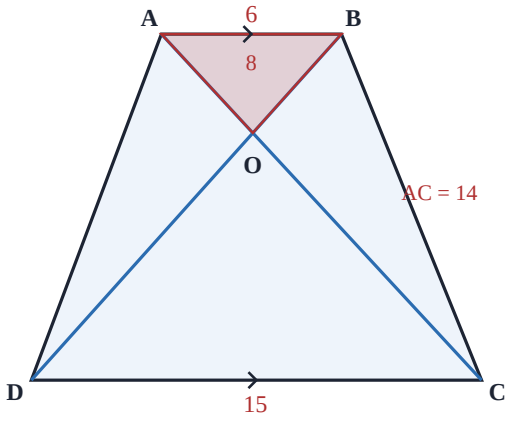
The hardest questions your teacher could ask on this unit: each one combines two or more skills. Show all your work. Answers are on the last pages.

Part 1 · Similar Triangles & Conversions

Question 1

[6 marks]

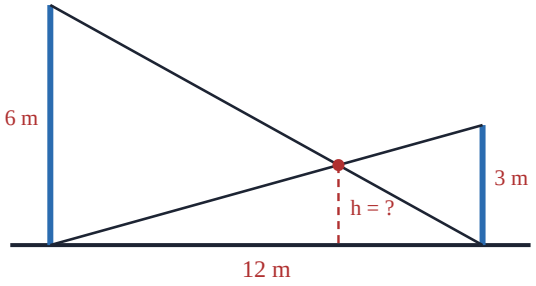
Trapezoid ABCD has AB ∥ DC, AB = 6 cm and DC = 15 cm. The diagonals meet at O, and AC = 14 cm. (a) Prove ΔAOB ~ ΔCOD. (b) Find AO and OC. (c) The area of ΔAOB is 8 cm². Find the areas of ΔCOD and ΔAOD.



Question 2

[5 marks]

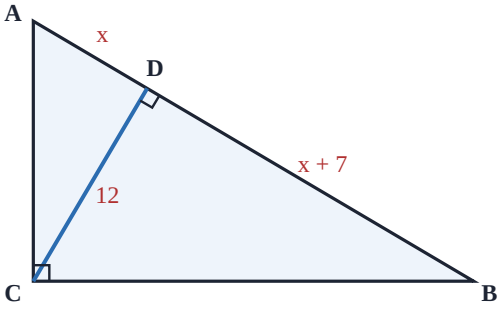
Two vertical poles 6 m and 3 m tall stand 12 m apart. A wire runs from the top of each pole to the bottom of the other. (a) How high above the ground do the wires cross? (b) How far from the taller pole is the crossing point? (c) Show that this height does not depend on the distance between the poles.



Question 3

[5 marks]

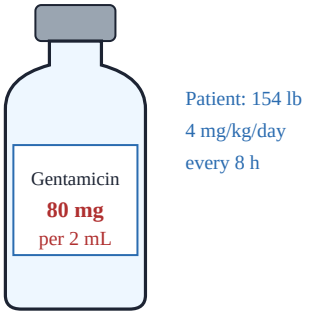
In right ΔABC (right angle at C), CD is the altitude to the hypotenuse AB. AD = x, DB = x + 7 and CD = 12. Find x, then AC, BC and the area of ΔABC.



Question 4

[4 marks]

A doctor orders Gentamicin at 4 mg per kg of body mass per day for a 154 lb patient, given in equal doses every 8 hours (1 kg = 2.2 lb). The vial label reads 80 mg in 2 mL. How many mL should each dose be, to the nearest hundredth?

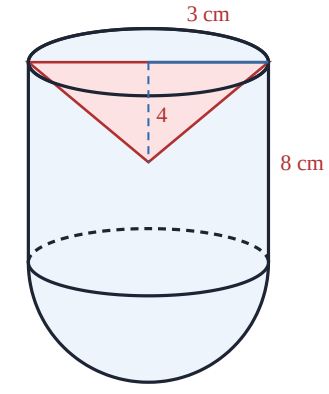


Part 2 · Surface Area & Volume

Question 5

[6 marks]

A solid cylinder (radius 3 cm, height 8 cm) has a cone of height 4 cm carved out of its top; the cone's base is the whole top face. A solid hemisphere is attached to the bottom. Find (a) the volume and (b) the total surface area, in terms of π and to the nearest tenth.

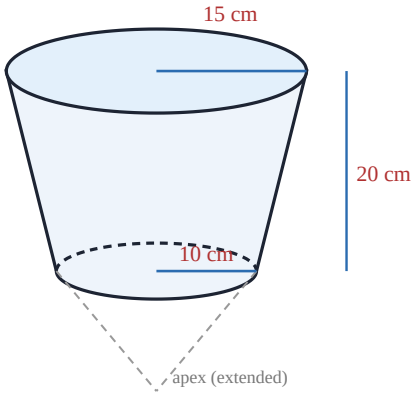


cone carved out · hemisphere added

Question 6

[6 marks]

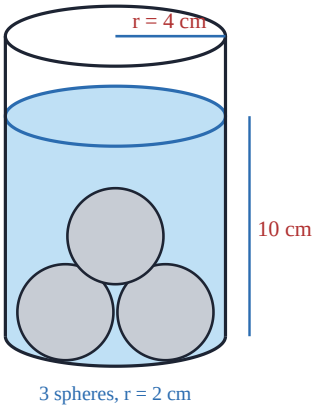
A bucket is a cone with its tip cut off: top radius 15 cm, bottom radius 10 cm, depth 20 cm. Use similar triangles to find the height of the full cone, then find the bucket's capacity in litres.



Question 7

[4 marks]

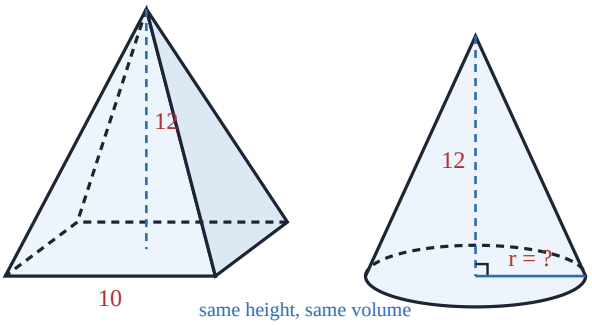
A cylindrical glass of radius 4 cm holds water 10 cm deep. Three metal spheres of radius 2 cm are dropped in and sink completely. How much does the water level rise, and what is the new depth?



Question 8

[6 marks]

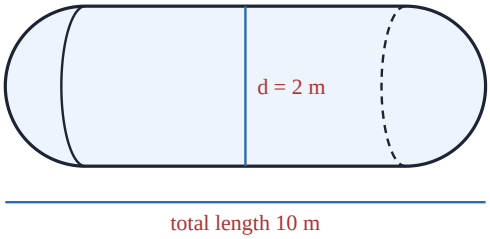
A square pyramid (base side 10 m) and a cone have the same height, 12 m, and the same volume. (a) Find the radius of the cone. (b) Which solid has the larger lateral (side) surface area, and by how much?



Question 9

[5 marks]

A fuel tank is a cylinder with a hemisphere on each end. Its total length is 10 m and its diameter is 2 m. Find (a) its outer surface area and (b) its capacity in kL.

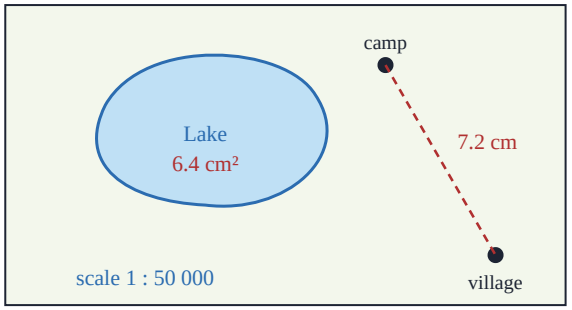


Part 3 · Scale & Scale Diagrams

Question 10

[5 marks]

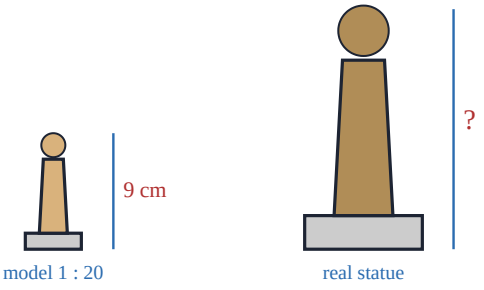
On a map with scale 1 : 50 000, a lake covers 6.4 cm² and the trail from the camp to the village measures 7.2 cm. (a) Find the real length of the trail in km. (b) How long does it take to walk at 4.8 km/h? (c) Find the real area of the lake in km² and in hectares.



Question 11

[5 marks]

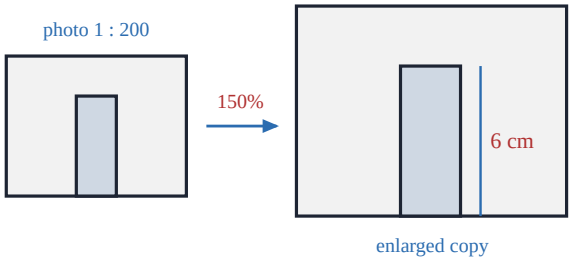
A clay model of a statue is built at a scale of 1 : 20. The model is 9 cm tall, has a surface area of 240 cm² and uses 150 cm³ of clay. Find the real statue's (a) height in m, (b) surface area in m² and (c) volume in m³.



Question 12

[4 marks]

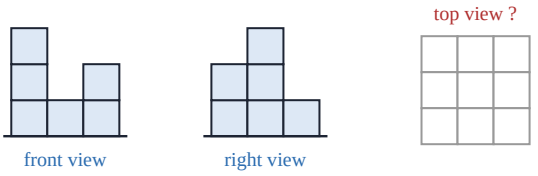
A photo of a building has a scale of 1 : 200. It is enlarged to 150% on a copier. (a) What is the scale of the enlarged copy? (b) The building is 6 cm tall on the copy. How tall is the real building?



Question 13

[5 marks]

A solid is built from 1-cm cubes. Its front view shows columns of 3, 1 and 2 cubes (left to right) and its right view shows 2, 3 and 1 cubes (front to back). The top view is unknown. (a) What is the greatest number of cubes the solid could have? (b) What is the least number? Explain.



ANSWER KEY

Part 1 · Similar Triangles & Conversions

Q1 (a) AB ∥ DC ⇒ ∠OAB = ∠OCD and ∠OBA = ∠ODC (alternate angles) ⇒ ΔAOB ~ ΔCOD (AA). (b) k = 15/6 = 2.5, so OC = 2.5·AO. AO + 2.5·AO = 14 ⇒ **AO = 4 cm, OC = 10 cm**. (c) Area COD = 8 × 2.5^2 = **50 cm^2**. ΔAOD and ΔAOB have the same height from A to line BD, and OD = 2.5·OB ⇒ area AOD = 8 × 2.5 = **20 cm^2**.

Q2 Let the crossing be h high and x from the taller pole. Wire from the short pole's top to the tall pole's foot: h/x = 3/12. Wire from the tall pole's top to the short pole's foot: h/(12 − x) = 6/12. So x = 4h and 12 − x = 2h ⇒ 6h = 12 ⇒ **h = 2 m**, **x = 8 m**. (c) For poles a and b, distance d: x = hd/b and d − x = hd/a ⇒ d = hd(1/a + 1/b) ⇒ **h = ab/(a + b)**, with no d. Check: 6·3/9 = 2 ✓

Q3 CD^2 = AD·DB ⇒ x(x + 7) = 144 ⇒ x^2 + 7x − 144 = 0 ⇒ (x + 16)(x − 9) = 0 ⇒ **x = 9** (a length is positive). AB = 25. AC^2 = 9·25 = 225 ⇒ **AC = 15**. BC^2 = 16·25 = 400 ⇒ **BC = 20**. Area = ½·15·20 = **150**.

Q4 154 ÷ 2.2 = 70 kg. Daily dose = 4 × 70 = 280 mg. Every 8 h = 3 doses ⇒ 93.33 mg per dose. Volume = 93.33 × 2/80 ≈ **2.33 mL**.

Part 2 · Surface Area & Volume

Q5 (a) V = π(9)(8) − ⅓π(9)(4) + ⅔π(27) = 72π − 12π + 18π = **78π ≈ 245.0 cm^3**. (b) Cone slant = √(3^2 + 4^2) = 5. SA = side 2π(3)(8) + cone π(3)(5) + hemisphere 2π(9) = 48π + 15π + 18π = **81π ≈ 254.5 cm^2** (no flat top is left).

Q6 H/(H − 20) = 15/10 ⇒ 15H − 300 = 10H ⇒ **H = 60 cm** (the removed tip is 40 cm tall). V = ⅓π(15^2)(60) − ⅓π(10^2)(40) = 4500π − 1333.3π ≈ 3166.7π ≈ 9948 cm^3 ≈ **9.95 L**.

Q7 Volume of spheres = 3 × (4/3)π(2^3) = 32π cm^3. Rise = 32π ÷ (π·4^2) = **2 cm**. New depth = **12 cm**.

Q8 (a) ⅓(10^2)(12) = ⅓πr^2(12) ⇒ r^2 = 100/π ⇒ **r ≈ 5.64 m**. (b) Pyramid slant = √(12^2 + 5^2) = 13 ⇒ lateral = 4·½·10·13 = 260 m^2. Cone slant = √(12^2 + 31.83) ≈ 13.26 ⇒ lateral = π(5.64)(13.26) ≈ 235.0 m^2. The **pyramid** is larger by about **25.0 m^2**.

Q9 r = 1 m; cylinder part = 10 − 2 = 8 m. (a) SA = 2π(1)(8) + 4π(1^2) = **20π ≈ 62.8 m^2** (two hemispheres = one sphere). (b) V = π(1^2)(8) + (4/3)π(1^3) = (28/3)π ≈ 29.3 m^3 = **29.3 kL**.

Part 3 · Scale & Scale Diagrams

Q10 (a) 7.2 × 50 000 = 360 000 cm = **3.6 km**. (b) 3.6 ÷ 4.8 = 0.75 h = **45 min**. (c) Area factor 50 000^2 = 2.5 × 10^9: 6.4 × 2.5 × 10^9 = 1.6 × 10^{10} cm^2 = 1.6 × 10^6 m^2 = **1.6 km^2 = 160 ha**.

Q11 (a) 9 × 20 = 180 cm = **1.8 m**. (b) 240 × 20^2 = 96 000 cm^2 = **9.6 m^2**. (c) 150 × 20^3 = 1 200 000 cm^3 = **1.2 m^3**.

Q12 (a) 1.5 : 200 = **3 : 400** (about 1 : 133.3). (b) 6 × 400/3 = 800 cm = **8 m**. Check: on the original photo it is 6 ÷ 1.5 = 4 cm, and 4 × 200 = 800 cm ✓

Q13 Use a 3 × 3 top grid: columns take the front heights (3, 1, 2), rows take the right-view heights (2, 3, 1). (a) Each square can hold at most min(column, row): row 2: 2 + 1 + 2 = 5; row 3: 3 + 1 + 2 = 6; row 1: 1 + 1 + 1 = 3 ⇒ **14 cubes**. (b) Each view height needs only one stack: put 3 where the 3-row meets the 3-column, 2 where the 2-row meets the 2-column, 1 where the 1-row meets the 1-column ⇒ 3 + 2 + 1 = **6 cubes**.